# Pramana Cx

TECHNICAL ARCHITECTURE & PRODUCT DOSSIER

TEAM PRAMANA

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DOCUMENT INDEX

# A4 technical dossier

This print-safe edition consolidates the current repository architecture, product requirements, technical requirements, data schema, application flow, design decisions, retrieval safeguards, release ledger and browser acceptance evidence.

01 / PURPOSE, USERS AND AUTHORITY

# What Pramana Cx is designed to decide

Pramana Cx is a governed evidence control plane for mission-critical EPC and data-centre commissioning. It connects controlled requirements to systems, assets, gates, tests, evidence, findings, schedules, shipments and immutable source evidence so an authorized engineer can make a defensible gate decision.

**Non-negotiable authority rule.** AI may extract, retrieve, rank, draft, classify and explain. Deterministic services calculate readiness, schedule feasibility and structured verdicts. Only an authorized human can accept a requirement or evidence item, approve a report, disposition a compliance finding or make a gate decision.

## Target users and friction removed

| User | Decision | How the application makes it easier |
|---|---|---|
| Commissioning manager / authority | Can a gate be defended today? | Readiness unifies accepted proof, stale/failed evidence, predecessors, blockers, owners and citations. |
| EPC / GC MEP package lead | What must be fixed before the next test? | Actions, Cx, evidence and schedule views preserve shared operational context. |
| QA/QC engineer | Does a controlled record meet a requirement? | Source-region provenance, deterministic checks and cited comparison proposals replace evidence hunting. |
| Planner / scheduler | What changed and should the baseline move? | Reviewed inputs, immutable CP-SAT versions and before/after explanations separate facts from speculation. |
| Owner / operations team | Is turnover complete and auditable? | Approved-gate-only manifests, hashes and independent verification make the package inspectable. |
| Field engineer / vendor | How can proof be captured safely on site? | Offline capture queues evidence idempotently until an authorized reviewer accepts it. |

## Application coverage

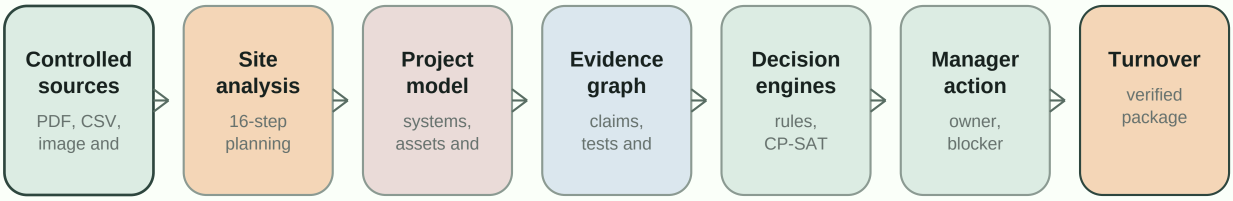
- Controlled sources, revisions, exact source-region citations and proposal review.
- Requirements, systems, assets, gates, evidence, findings, readiness and turnover packs.
- Schedule inputs, CP-SAT baselines, immutable versions, live events and advisory risks.
- Cx standards, cited checklists, deterministic readings, reports and human approval.
- Sixteen-stage Site Analysis with review, deterministic conflicts, planning insights and versioned finalization.
- Readiness, commissioning plans, controlled tests, evidence, findings and turnover linked to accepted project records.
- USD-authoritative Financial Modeler with INR display, NPV, IRR, payback, sensitivity and key-driver explanations.
- Site-derived procurement, approved shipment materialization, maps, routes, weather and schedule events.
- RackDB-style digital rack revisions, project-derived capacity, custom GPU/rack components, wiring and controlled model exports.
- Compliance, controlled RAG, knowledge/RFI intelligence, graph, changes and Command Center.

02 / TEAM, PROPOSITION AND CONNECTED OUTCOME

# Three builders, one governed project story

**Atharva Deo, Bhavik Sheth and Dravvya Jain** are Team Pramana: a three-member product and engineering team building decision infrastructure for mission-critical data-centre delivery. Pramana Cx is not another document chatbot. It connects the work a project manager must already coordinate—site feasibility, requirements, systems, evidence, tests, compliance, logistics, economics and turnover—into one explainable control plane.

CONNECTED MANAGEMENT LOOP




## Core USPs

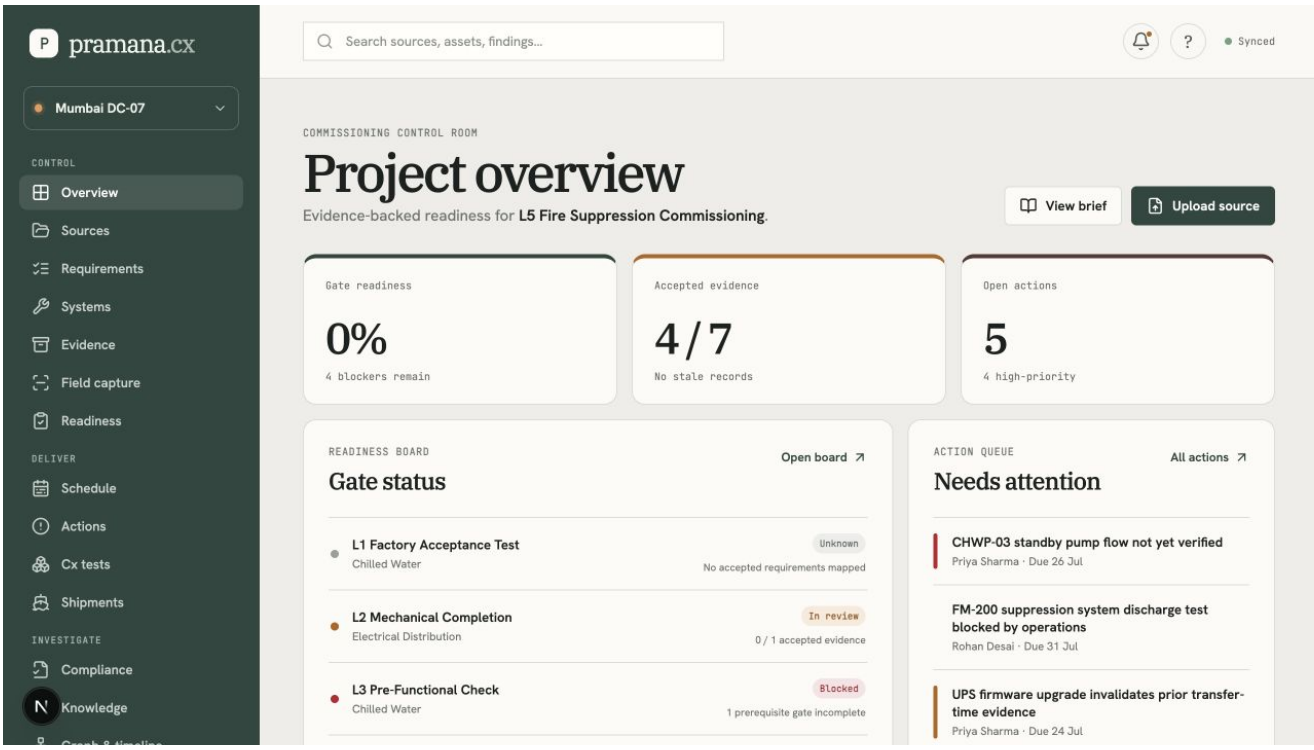
| Capability | What has been built | Why it is defensible |
|---|---|---|
| Controlled RAG | Immutable regions, full-text and vector candidates, BGE reranking and cited synthesis. | Scope is filtered before ranking; unsupported claims are removed. |
| Site Analysis | Sixteen planning stages, evidence-aware defaults, deterministic rules, review and versioned insights. | Assumptions, confirmations, conflicts and missing decisions stay separate. |
| Readiness + Cx | Requirements and evidence generate context, procedures, blockers, progress and gate decisions. | Only accepted proof affects deterministic readiness. |
| Local Gemma | Bounded extraction, classification, drafting and manager-facing explanation. | Model output stays proposed and records provider provenance. |
| Economics | Canonical USD model with INR display, NPV, IRR, payback, sensitivity and key drivers. | Formula results persist independently of narrative or display currency. |
| Supply + routes | Site-derived plans, approval, shipments, maps, weather and schedule events. | Live, simulated, fallback and unavailable states are explicit. |
| Evidence + compliance | Field/PDF/image intake, claim links, exact citations, relevance gates and disposition. | AI cannot certify, self-approve or compare unrelated concepts. |
| Digital racks | Project-derived racks, custom GPUs/components, wiring, revisions and model exports. | Objects remain tied to project scope and revision. |

03 / APPLICATION EXPERIENCE

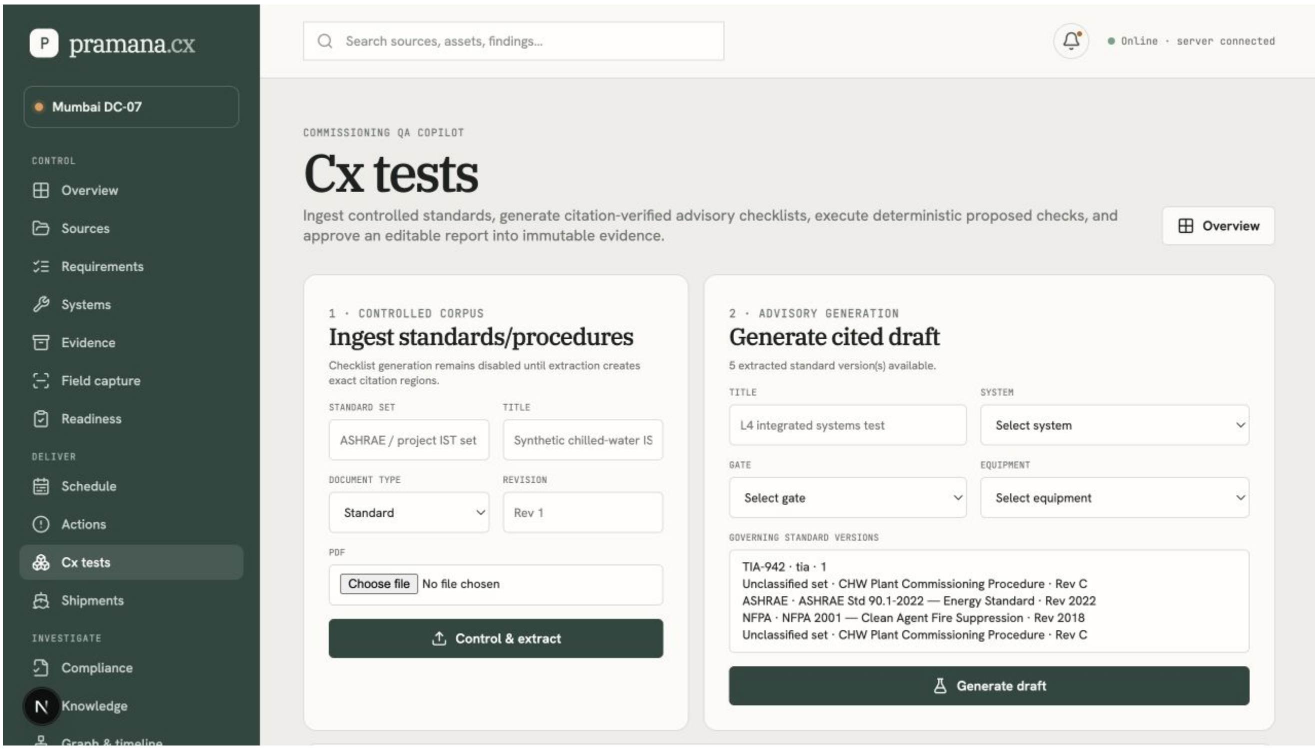
# Task-oriented, cited and resilient

The workspace groups navigation as Control, Deliver and Investigate. Deep links preserve the selected gate, task, finding or shipment across feature boundaries. Busy-state and progress behaviour is operation-specific, bounded and visible, so long-running work cannot silently replace another action.

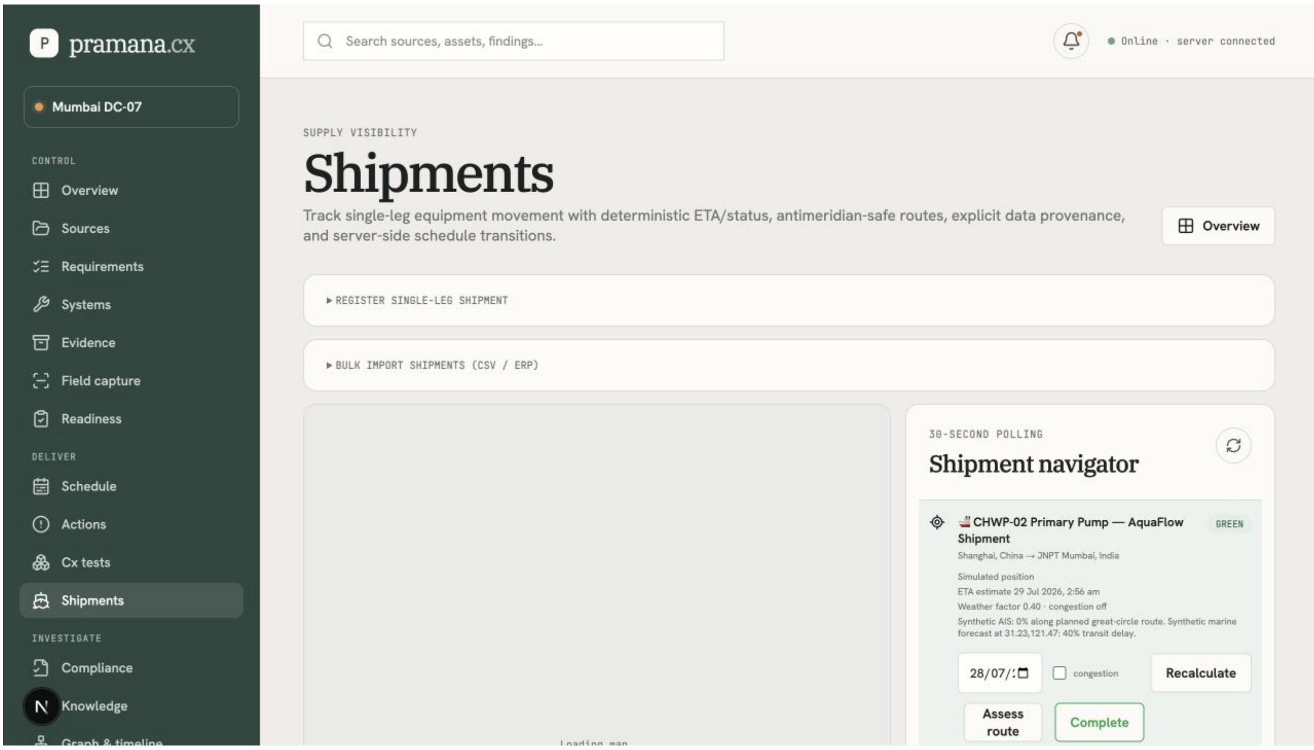
## Live browser-captured application surfaces



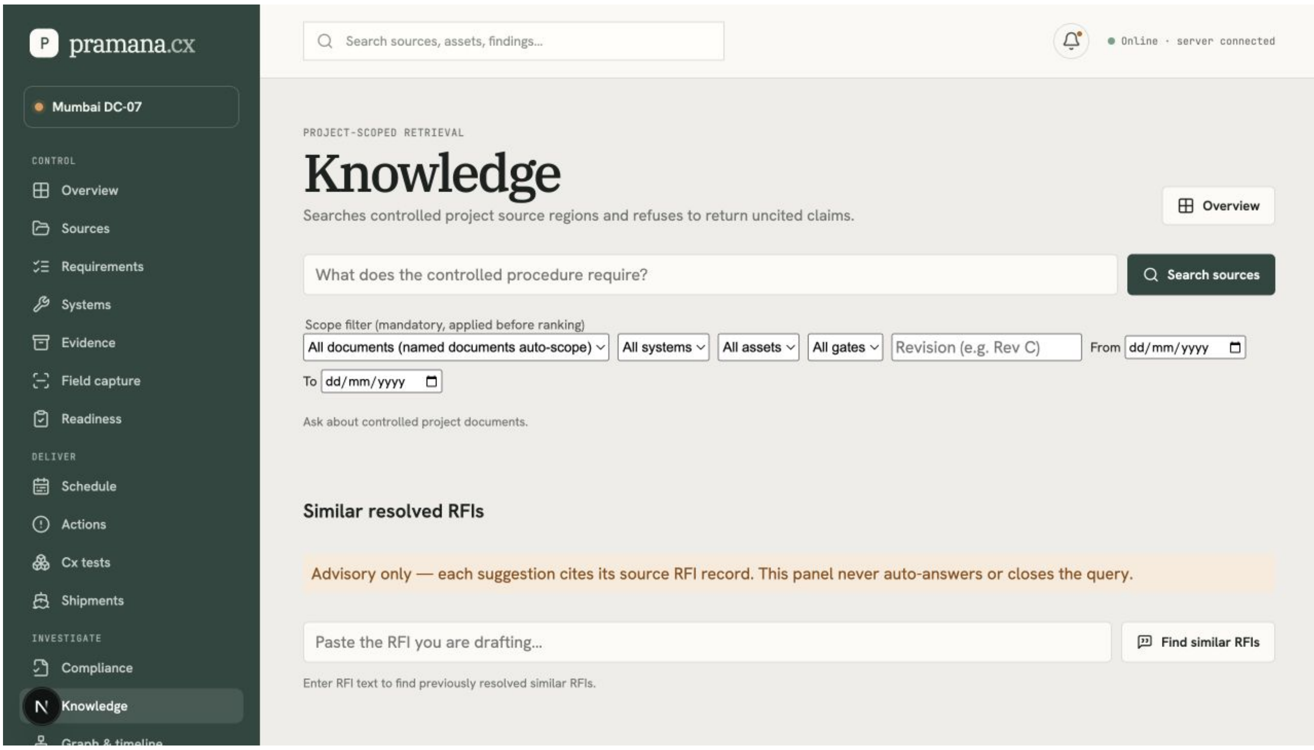

**Overview and grouped navigation - one project context and a complete task map.**




**Governed Cx tests - cited standards, deterministic readings, human review and report approval.**




Shipment visibility - route/position context, ETA provenance and controlled operational status.




Cited knowledge intelligence - project-scoped retrieval and source-grounded answers.

## End-to-end user journey

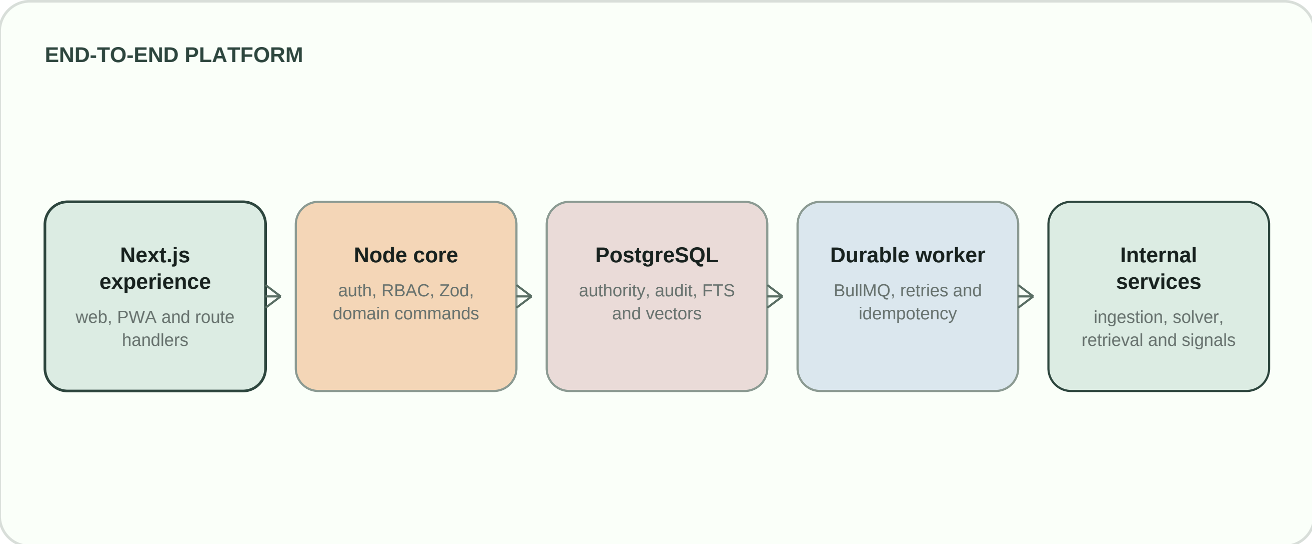
| Step | Surface | Data flow | Outcome |
|---|---|---|---|
| 1 | Overview / Settings | Project membership scopes all loads and actions. | User enters an authorized project context. |
| 2 | Sources / Cx | Bytes -> hash/version -> source regions -> durable extraction. | Reviewer sees cited proposals. |
| 3 | Systems / Requirements | Accepted records create typed system/asset/gate relationships. | Facility context is traversable. |

| Step | Surface | Data flow | Outcome |
| --- | --- | --- | --- |
| 4 | Evidence / Field / Cx | Evidence starts pending; review adds acceptance/audit state. | Proof contributes to readiness only when accepted. |
| 5 | Schedule / Shipments | Validated event -> delta detector -> CP-SAT only if material. | New immutable version or status-only update. |
| 6 | Actions / Compliance / Knowledge | Findings, alerts and citations retain record identity. | Owner resolves human-reviewed work. |
| 7 | Readiness / Turnover | Rules + authorized decision -> manifest. | Approved gates produce verifiable turnover. |

03 / SYSTEM ARCHITECTURE

# Modular monolith with isolated services

The Next.js core is the authenticated, project-scoped front door and business authority. Durable or expensive work passes through Redis/BullMQ. PostgreSQL is the transactional source of truth; object storage holds immutable bytes; typed relational edges provide governed graph traversal.




## Service ownership and boundaries

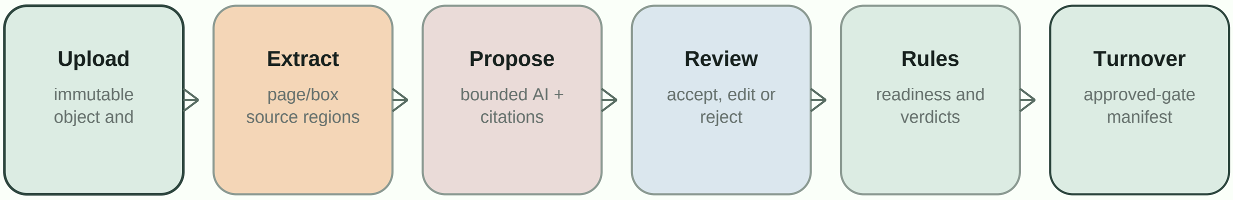
| Component | Owns | Must not own |
|---|---|---|
| Next.js core | Authentication, authorization, validation, domain commands, audit and project scoping. | Long solver/extraction work or vendor-specific direct AI calls. |
| Worker + BullMQ | Retryable background operations, stable job identity and durable job state. | Interactive authorization or final human approval. |
| Ingestion service | Format extraction into structured regions. | Business authority, readiness or permission decisions. |
| CP-SAT solver | Feasible schedule or explicit bottleneck/infeasibility result. | Database/object storage access, approvals or LLM calls. |
| Retrieval service | Embeddings and cross-encoder reranking. | Database credentials, tenant decisions or final-answer authority. |

**Architecture decision.** CP-SAT runs as a stateless dedicated service because an optimization run must not block the Node request process. The caller passes the complete DAG/constraints; the solver does not read Postgres or object storage.

04 / DATA FLOW, USER FLOW AND NAVIGATION

# From controlled source to approved turnover

AUTHORITY AND EVIDENCE FLOW

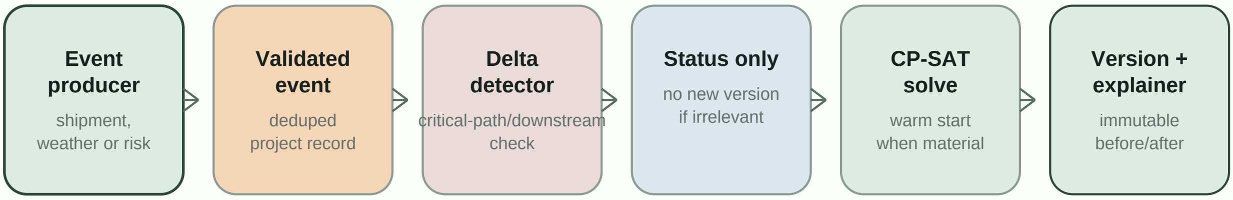



## Runtime job lifecycle

- The browser submits a project-scoped command. The core authenticates, authorizes, validates and persists idempotency/audit state.
- Long-running work becomes a stable BullMQ job. The caller receives a job reference and can read progress without freezing the UI.
- The worker reads immutable input, invokes an internal service under timeout/retry rules, validates structured output and persists result, edges and audit event.
- The UI reads the authoritative result and labels advisory versus authoritative state.

## Schedule event flow

MATERIALITY-GATED SCHEDULE HANDLING




**Schedule safety boundary.** Predictive risk can observe, explain and propose mitigation options. It cannot apply a mitigation, alter dependencies/resources, invoke the solver autonomously or change dates.

## Navigation contract

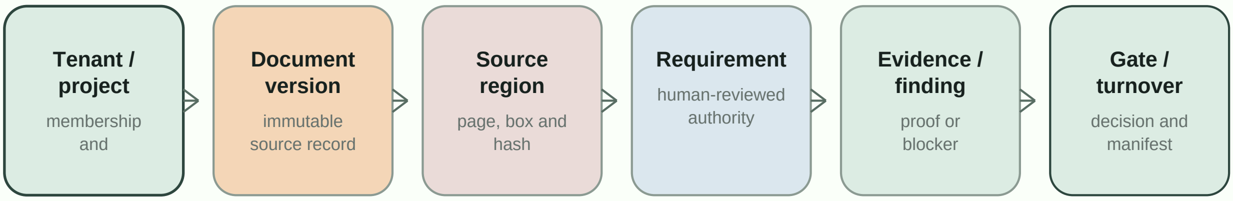
Cross-feature actions include a stable identifier in the URL, for example `/readiness?gate=...`, `/schedule?task=...`, `/actions?finding=...` and `/shipments?shipment=...`. The destination validates, consumes and focuses the exact record; a missing target degrades safely rather than producing an unrelated generic screen.

05 / DATABASE, PROVENANCE AND SCHEMA

# Transactional authority, immutable bytes, traversable relationships

Every authoritative record is tenant and project scoped. Cross-project identifiers are rejected at the API boundary. Normalized relational tables remain the authority; the `edges` relation enables governed traversal; immutable object storage retains original source/evidence/report/turnover bytes.

CORE ENTITY PATH




## Persistence domains

| Domain | Representative records | Integrity intent |
|---|---|---|
| Identity and scope | tenants, users, projects, project_members, sessions | Membership is evaluated server-side for every record and action. |
| Sources and knowledge | documents, document_versions, source_regions, knowledge_chunks | Immutable bytes, exact citations and embedding-model tags prevent mixed vector spaces. |
| Evidence control | requirements, evidence, systems, assets, gates, findings, decisions, turnover packs | Human review and deterministic readiness govern state transitions. |
| Planning and supply | schedule_tasks, resources, schedule_versions, schedule_events, risks, shipments | Accepted inputs, immutable solve history, event deduplication and provenance. |
| Cross-cutting | edges, durable_jobs, alerts, audit_events | Typed relationships, retry/idempotency and append-only hash-linked history. |

## Provenance safeguards

- Immutable SHA-256 object record is stored before extraction; source regions retain document version, page, bounding box and content hash.
- New revisions never overwrite old regions; superseded, rejected, failed or unauthorized sources cannot be retrieval candidates.
- Accepted-only evidence can prove a requirement. Stale/failed evidence invalidates readiness; edits trigger audited recalculation.

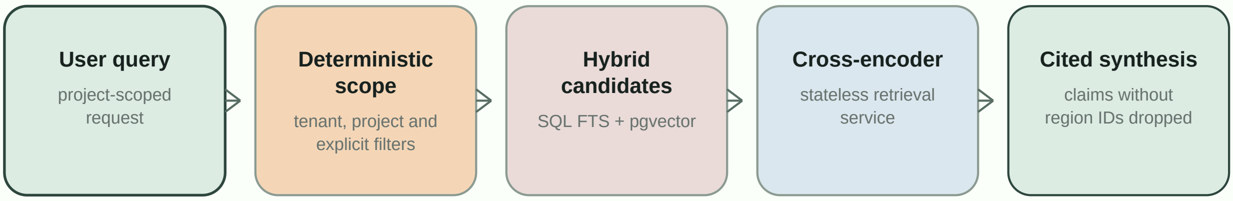
- Audit events are append-only and hash-linked per project.

06 / AI, RAG AND AGENT BOUNDARIES

# Bounded, cited, time-limited and non-authoritative

Core generation and embedding calls resolve through a central provider boundary. Local Ollama is the intended deployment default: gemma4:e2b for structured generation and nomic-embed-text:latest for 768-dimensional embeddings. Gemini and NVIDIA NIM can be selected explicitly. The deterministic mock provider is verification-only and must never be deployed.

CONTROLLED RAG PIPELINE




## Provider and RAG safeguards

- Bounded prompt/context sizes, output token caps, request deadlines, schema validation and retry limits.
- Client aborts, per-operation busy state and progress feedback prevent frontend blocking.
- Tenant/project and supplied metadata filters are enforced in SQL before ranking, not after retrieval.
- Only explicit reviewer/user filters and deterministic title routing may narrow authority-bearing scope. The LLM cannot choose system, asset, gate, revision, type or document scope.
- Groundedness is code-enforced: an answer claim without a resolvable source-region identifier is dropped.

## Five operational agent capabilities

| Capability | Implementation posture | Non-negotiable boundary |
|---|---|---|
| Commissioning QA Copilot | Cited standards, checklist proposal, deterministic readings, human-routed narrative review and draft report workflow. | No automatic approval; approved report only becomes immutable evidence after review. |
| Supply Chain Visibility and Risk | Shipment legs, multi-mode map, polling and weather/congestion/AIS assessment. | Delayed/recovered event is provenance-labelled and cannot directly alter a schedule. |
| Specification and Quality Compliance | Semantic candidate discovery plus deterministic numeric/boolean/category comparisons and precedent handling. | Creates proposed findings only; an engineer decides disposition. |
| Predictive Schedule Risk | Bounded observations of procurement, lead-time, workforce and weather with materiality/deduplication. | Advisory risks/mitigations only; it cannot change dates or apply an option. |
| Knowledge and RFI Intelligence | Hybrid project-scoped retrieval, graph context, reranking, cited synthesis and similar-RFI retrieval. | Returns grounded citations or explicit no-results; never invents authority. |

07 / TECHNOLOGY, SDKS, APIS AND SERVICES

# Concrete technology inventory

| Layer | Technology | Purpose |
| --- | --- | --- |
| Web | Next.js 16.2.10, React 19, TypeScript 5.7 | Server-rendered application, route handlers, typed UI and responsive PWA surfaces. |
| Schema and validation | Drizzle ORM 0.45, Drizzle Kit 0.31, Zod 3.24 | SQL migrations, relational authority and validated contracts. |
| Database | PostgreSQL 16 target topology, pgvector, Postgres FTS | Transactional state, constraints, audit/jobs and hybrid search. |
| Queue/cache | Redis, BullMQ 5.80, ioredis 5.11 | Durable background work, idempotency coordination and rate limits. |
| Storage | Filesystem locally; MinIO/S3 via AWS SDK v3 | Immutable original, report and turnover artifacts with signed reads. |
| Extraction / solver / retrieval | Python/FastAPI, PyMuPDF, Google OR-Tools CP-SAT, BAAI bge models | Controlled source extraction, optimization, embeddings and reranking. |
| AI providers | Ollama gemma4:e2b and nomic-embed-text; optional Gemini and NVIDIA NIM | Explicit schema-validated advisory provider choices. |
| Maps and 3D | Leaflet, React Leaflet, Turf, searoute-js, OpenStreetMap, Three.js 0.185 | Route geometry, weather-aware logistics and project rack/model rendering. |
| Data and exports | PapaParse 5.6, csv-parse 7, PDFKit 0.19 | Site-input CSV ingestion and controlled PDF/CSV/vendor/model deliverables. |
| Identity | bcryptjs, OTPAuth, optional Clerk Next.js SDK | Owned credentials/TOTP or external identity adapter with app-owned RBAC. |

## API group contract

| Group | Representative routes | Expectation |
| --- | --- | --- |
| Identity | /api/auth/*, /api/profile | Credentials/Clerk, sessions, TOTP and rate limits. |
| Project authority | /api/projects, /members, /activate, /alerts, /site-analysis | Membership, planning snapshots, finalization and role enforcement. |
| Evidence control | /sources, /evidence, /requirements, /gates, /turnover-packs | Versioning, citations, review and readiness. |
| Schedule and risk | /schedule/tasks, /resources, /baseline, /events, /risks, /shipment-plans | Reviewed input, immutable versions, approval and solver isolation. |
| Economics and models | /financial-model, /rack-model, /technology-drafts | Versioned scenarios, rack/GPU revisions and controlled vendor packages. |
| Cx/compliance/knowledge | /cx/*, /compliance/*, /knowledge/*, /graph | Cited, scoped, proposed-only advisory flows. |

08 / PREREQUISITES, INSTALLATION AND CONFIGURATION

# Local topology and production configuration

Local development can use separate PostgreSQL, Redis, ingestion and solver processes or the complete Compose topology. Production must use secret-managed configuration, non-degraded infrastructure and a model endpoint reachable from both web and worker services.

## Installation

```
npm install
cp .env.example .env.local
npm run db:migrate
npm run db:seed
npm run system:start

npm run typecheck
npm run build
npm run verify:all

ollama pull gemma4:e2b
ollama pull nomic-embed-text:latest
MODEL_PROVIDER=ollama EMBEDDING_PROVIDER=ollama npm run verify:ollama
```

## Environment and infrastructure requirements

| Requirement | Local | Production |
|---|---|---|
| Runtime | Node 22, npm, Python 3; Docker Desktop for Compose. | Pinned images/runtimes; separate web and worker processes. |
| Database | Reachable Postgres; migrations applied; pgvector where used. | TLS, backups, restore rehearsal, role separation and empty-db migration proof. |
| Secrets | .env.local never committed. | Secret manager, rotation policy and no secrets in artifacts/logs. |
| Object storage | Filesystem or local MinIO. | Private bucket, encryption, lifecycle/retention and signed URL policy. |
| Models | Optional local Ollama for real smoke tests. | Reachable model service for web/worker with data-processing approval. |

## Key configuration groups

| Group | Representative keys | Release condition |
|---|---|---|
| Auth | APP_BASE_URL, AUTH_MODE, AUTH_ENCRYPTION_KEY, Clerk keys | Use credentials or Clerk, never development mode. |
| AI/RAG | MODEL_PROVIDER, EMBEDDING_PROVIDER, OLLAMA_BASE_URL, model keys, GEMINI_API_KEY, NIM_API_KEY | No mock provider; endpoints reachable and bounded. |
| Infrastructure | Database/Redis URLs, object storage settings, INFRA_ALLOW_DEGRADED | All dependencies healthy; degraded mode false. |
| Signals | Procurement, lead-time, workforce, weather, AIS/congestion settings | Live/synthetic/unavailable provenance is explicit. |

**Installation exit criteria.** Migrations pass, dependencies are healthy, an authorized source can be processed, the verification matrix passes and the real-provider smoke test is run separately when a provider is configured.

09 / TESTING, ERROR HISTORY AND MITIGATION

# Release confidence is a matrix, not a green build

| Area | Coverage | Evidence |
|---|---|---|
| Build/migration | TypeScript, optimized build, Drizzle checks and migration replay. | Valid artifact and schema state. |
| Security/tenancy | Credentials/TOTP, rate limits, cross-project rejection and audit chain. | Server-side enforcement and append-only history. |
| Evidence to turnover | Pending/accepted proof, readiness, gate decision and manifest verification. | Accepted-only proof and independently verifiable turnover. |
| Schedule/risk | Inputs, dependencies, CP-SAT feasibility, immutable versions and recovery. | No solve for irrelevant event; explicit degradation. |
| Cx/compliance/knowledge | Citations, deterministic readings, review, filters, reranking and groundedness. | No ungrounded or unauthorized decision. |
| Browser acceptance | Sidebar journeys, mobile/desktop layouts, busy state, maps and console capture. | No duplicate-key warning, error boundary or overflow. |

## Cross-branch and application mitigation ledger

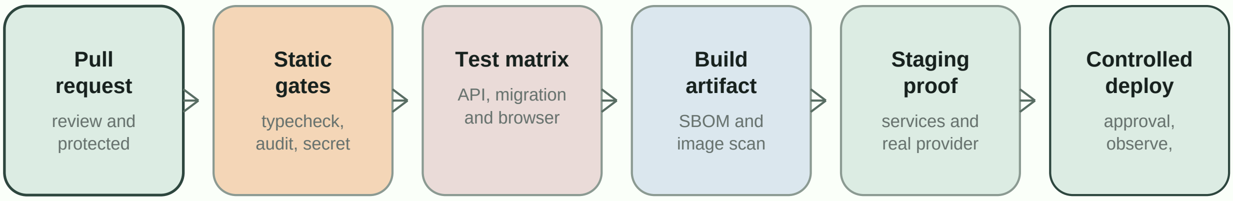
| Failure found | Mitigation now built |
|---|---|
| Mock TypeScript output could impersonate an LLM. | Explicit generation/embedding providers, Ollama default, mock test-only rule and provenance. |
| Direct Gemini/provider bypasses produced inconsistent policy. | Central schema-validated provider boundary and real-provider smoke tests. |
| Long prompts made operations look stuck. | Prompt/token/request bounds, client abort, progress, busy state and bounded retry. |
| Deep links lost record context. | Destination validation and focus contracts for readiness, schedule, actions and shipments. |
| Duplicate React keys and business rows. | Stable composite UI keys, unique business constraints and integrity verification. |
| Conflicting branch migrations. | Final-schema-first reconciliation, ordered migrations and replay validation. |
| Processed documents were absent/misattributed in RAG. | Extraction-indexing contract, backfill, SQL document filtering and citation validation. |
| Map fallbacks looked operational. | Coordinate checks, HTTPS/deadline routing, mode-aware provenance and explicit fallback notice. |
| Raw JSON/hashes/synthetic cues degraded UX. | Presentation helpers, human labels, bounded formatting and source-state labels. |
| Architecture documentation regressed through merge work. | Visible README diagrams, this PDF dossier, release ledger and documentation-first review. |

**Local verification result.** Type check, optimized production build, production dependency audit and the post-fix local verification matrix have passed. Target-environment infrastructure, real provider acceptance, accessibility and load validation remain production gates.

10 / DEPLOYMENT READINESS AND FUTURE CI/CD

# Controlled deployment, not an environment-variable switch

FUTURE CI/CD PIPELINE




## Required production gates

- Secrets and identity: production AUTH_MODE, encryption/session settings, base URL and identity credentials are secret-managed and validated.
- AI: selected Ollama, Gemini or NIM generation/embedding endpoint is reachable by web and worker; deadline, structured output and embeddings are smoke-tested in release configuration.
- Data plane: pgvector is available; migrations-from-empty, S3/MinIO lifecycle, Redis/worker recovery and backup/restore are rehearsed.
- Internal services: ingestion, solver and retrieval health checks are green; TLS/internal network restrictions and non-degraded infrastructure are active.
- Live signals: AIS/weather/congestion/procurement provenance and failure semantics are validated; synthetic paths are disabled or visibly labelled outside test/demo environments.
- Operations: CI, structured logs/correlation IDs, metrics, traces, queue dashboards, alerting, incident ownership, accessibility and load testing are present before release.

## Explicit non-claims

> **Engineering authority remains human.** Pramana Cx assists engineering judgement. It does not independently certify a facility, replace the commissioning authority or engineer of record, issue Tier/TIA/BICSI/statutory approval, autonomously close findings/waivers or represent simulated operational feeds as live evidence.

## Maintenance references

**README.md** - implementation map, diagrams, defect audit and configuration. **STATUS.md** - authoritative shipped/open ledger. **PLANNER/PRD.md** and **PLANNER/TRD.md** - product scope and technical contracts. **PLANNER/Schema.md**, **AppFlow.md**, **DesignDecisions.md** and **RetrievalArchitecture.md** - data model, flows, decisions and retrieval safeguards.

APPENDIX A / COMPLETE COMPONENT ARCHITECTURE

# Full component and responsibility map

This appendix records the implementation-level architecture omitted by an executive summary. It identifies every major runtime boundary, its state, its caller and the rule that prevents it from becoming an unauthorized decision-maker.

| Layer | Components | Responsibility | Hard boundary |
|---|---|---|---|
| Experience | Next.js server-rendered pages; responsive browser shell; mobile navigation; field-capture PWA; Three.js 404. | Project workbenches for Overview, Sources, Requirements, Systems, Evidence, Field Capture, Readiness, Schedule, Actions, Cx, Shipments, Compliance, Knowledge, Graph, Command Center, Changes and Turnover. | UI only requests project-scoped APIs; it does not make authority decisions locally. |
| Core API | Next.js route handlers; Zod contracts; permission middleware; model-provider facade; presentation helpers. | Authenticates, authorizes, validates, persists domain commands, returns read models and controls provider selection. | No direct unscoped data access or direct vendor-model bypass. |
| Business authority | Requirements/evidence/gates/findings/decisions; readiness rules; Cx deterministic verdicts; audit chain; typed edges. | Makes relational state authoritative and invokes deterministic calculations over accepted records. | LLM output has proposed state only; no AI can mark a gate ready. |
| Durable execution | Redis, BullMQ, durable_jobs, idempotency_records, worker and bounded retry wrappers. | Handles parse, embed, solve, polling, report and other retryable work with stable job identity. | Jobs cannot bypass review/permission checks; retries are bounded and observable. |
| Ingestion | PyMuPDF service on 8001; PDF/CSV/XLSX parser; source-region extraction; object-storage boundary. | Creates page/box/row/cell provenance and structured extraction output from immutable source versions. | Never becomes a source of business truth without core validation and human review. |
| Scheduling | FastAPI OR-Tools CP-SAT service on 8002; accepted task/resource inputs; delta detector; immutable versions; explainer. | Solves feasibility/optimization and returns explicit result or bottleneck report; carries completed work as fixed input on warm start. | Solver is stateless, has no database credentials and never changes a date without a reviewed command. |
| Retrieval | Postgres FTS + pgvector; knowledge_chunks; graph context expansion; retrieval service on 8003; cross-encoder reranker. | Retrieves project-scoped controlled regions and ranks candidates before cited synthesis. | Mandatory tenant/project/metadata filters happen before rank; service has no database credentials. |
| Operational signals | Shipments, AIS/weather/congestion, procurement/lead-time/workforce, schedule_events, risks and alerts. | Persists observations, calculates materiality/deduplication and exposes advisory impact. | All sources explicitly label live, synthetic or unavailable; no signal directly changes readiness/schedule. |
| Object and security | PostgreSQL, pgvector, local/MinIO/S3 objects, signed URLs, credentials/TOTP or Clerk, rate limits. | Owns transactional state, immutable artifacts and controlled access. | Cross-project IDs rejected; audit is append-only/hash-linked; production secrets are external. |

## Full agent event contract

| Event | Producer | Trigger | Persisted impact |
|---|---|---|---|
| TEST_FAILED | Cx workflow | Deterministic numeric/threshold or boolean/presence proposed failure only. | Finding/NCR proposal and blocked gate flow; narrative observations stay human review. |
| shipment_delayed | Shipment assessment | Status transition into at-risk/delayed, never a repeated poll state. | One event per affected task fan-out; delta detector assesses schedule impact. |
| shipment_recovered | Shipment assessment | Prior delay recovers to on-time. | Clears stale active alert and may enter the same delta assessment. |

| Event | Producer | Trigger | Persisted impact |
|---|---|---|---|
| predicted_risk_delay | Predictive risk poll | Task + risk-type materiality threshold crosses or changes materially. | Advisory risk/mitigation event; no direct re-solve or schedule mutation. |

APPENDIX B / ARCHITECTURE DECISION RECORDS

# All committed architecture decisions

| ID | Decision | Implementation consequence |
|---|---|---|
| ADR-001 | Local modular monolith | One typed core for MVP authority; isolated services only for extraction, solve and retrieval. |
| ADR-002 | Postgres + Drizzle | Normalized relational source of truth with migrations and constrained traversal. |
| ADR-003 | Object-store bytes | Immutable originals and exports live behind one local/S3-compatible boundary. |
| ADR-004 | Hybrid lexical + semantic retrieval | Exact FTS plus project-scoped vector candidates and reranking. |
| ADR-005 | Durable asynchronous ingestion | Long work runs through retryable jobs and human checkpoints. |
| ADR-006 | Deterministic readiness | AI can propose; rules and authorized humans decide authority-bearing state. |
| ADR-007 | Project-scoped RBAC + strong approval | Server-side membership with TOTP gate protection. |
| ADR-008 | Offline-capable field PWA | Idempotent local queue supports site capture without authorizing offline decisions. |
| ADR-009 | Dedicated CP-SAT service | Official OR-Tools backend avoids unstable WASM and Node event-loop blocking. |
| ADR-010 | ModelProvider for schedule extraction | Ambiguous extracted tasks/resources require human review. |
| ADR-011 | Deterministic solver objective order | Deadline overrun first, idle time second; never delegated to an LLM. |
| ADR-012 | Immutable schedule versions | No in-place schedule mutation; every solve produces inspectable history. |
| ADR-013 | Delta-gated re-solves | Only critical/downstream impact invokes CP-SAT. |
| ADR-014 | Internal agent-service pattern | Domain services integrate through the same controlled boundary as solver. |
| ADR-015 | Pinned event contract | Event payload/dedup semantics are defined while a general orchestrator remains deferred. |
| ADR-016 | Cx proposed verdict + draft report | Narrative remains human-routed; report must be approved before evidence authority. |
| ADR-017 | Leaflet shipment navigator | Route geometry, click-to-zoom and explicit live/simulated labels. |
| ADR-018 | LLM confined to drafting | Acceptance/shipment classification remain deterministic or human reviewed. |
| ADR-019 | Compliance proposed-findings queue | Modality-tiered flags and clause-vs-line evidence diffs require disposition. |
| ADR-020 | Predictive risk as advisory poll | Dedicated live-events/risk surfaces cross-link critical path but do not reschedule. |
| ADR-021 | Scoped knowledge + edges graph | User-facing cited RAG and graph/timeline reuse project data, no parallel graph authority. |
| ADR-022 | Command Center as read/cross-link surface | Unified deduplicated alerts, recovery clearing and no direct state mutation. |

APPENDIX C / COMPLETE DATABASE INVENTORY

# Schema domains and real table inventory

The current Drizzle schema defines 45 tables. The following inventory preserves every authoritative domain rather than reducing the database to a generic ER picture.

| Domain | Tables | Authority and relationship role |
|---|---|---|
| Scope and identity | tenants, users, auth_sessions, projects, project_members | Tenant/project ownership, membership, sessions and RBAC. |
| Object/source control | storage_objects, documents, document_versions, source_regions | Immutable objects, revision lineage and exact extraction provenance. |
| Facility and authority | systems, assets, gates, requirements, evidence, findings, decisions | Controlled requirements, proof, blockers and authorized decisions. |
| Graph and audit | edges, audit_events, durable_jobs, idempotency_records, alerts | Typed traversal, hash chain, job lifecycle, dedupe and alert state. |
| Commissioning | cx_checklists, cx_checklist_steps, cx_clause_citations, cx_test_records, cx_step_results | Cited checklist execution, deterministic readings and report evidence. |
| Compliance and knowledge | compliance_precedents, compliance_checks, knowledge_chunks | Proposed deviations, exact precedents, model-tagged hybrid retrieval units. |
| Logistics/turnover | shipments, turnover_packs | Asset-linked delivery observations and approved manifest artifacts. |
| Scheduling | schedule_tasks, schedule_resources, schedule_task_resources, schedule_dependencies, schedule_versions, schedule_assignments | Reviewed DAG/capacity inputs, immutable versions and assignments. |
| Risk and learning | schedule_events, risk_signals, schedule_risks, teachback_notes | Validated event history, source observations, advisory risks and reviewed teach-back. |

## Relationship rules

- Tenant owns projects; project membership authorizes every project-scoped command and read.
- Document versions own source regions; source regions ground requirements, evidence, Cx citations, knowledge chunks and compliance comparisons.
- Systems contain assets; gates govern requirements; evidence can prove requirements and affect assets; findings block gates.
- Edges represent typed relationships such as PROVES, AFFECTS, REQUIRES and PRECEDES, but normalized tables remain business authority.
- Schedule versions contain assignments; tasks have dependencies/resources and are observed by risk signals/events without folding advisory risk into readiness.
- Business-key uniqueness, exact-row reconciliation and relational integrity checks protect merged/seeded data from duplicate identity corruption.

APPENDIX D / COMPLETE API ENDPOINT CATALOG

# All route-handler surfaces

The following catalog is generated from the current `src/app/api/**/route.ts` tree at PDF-build time. Dynamic route segments remain visible so contract owners can identify the entity scope.

| Group | Endpoint |
|---|---|
| login | /apiauth/login |
| logout | /apiauth/logout |
| password | /apiauth/password |
| register | /apiauth/register |
| sessions | /apiauth/sessions/[sessionId] |
| totp | /apiauth/totp/challenge |
| totp | /apiauth/totp/disable |
| totp | /apiauth/totp/enroll |
| totp | /apiauth/totp/verify |
| checks | /apicompliance/checks/[checkId]/review |
| precedents | /apicompliance/precedents/[precedentId]/review |
| conversations | /apicopilot/conversations/[conversationId]/attachments |
| conversations | /apicopilot/conversations/[conversationId]/messages |
| conversations | /apicopilot/conversations |
| memories | /apicopilot/memories |
| checklists | /apicx/checklists/[checklistId]/review |
| checklists | /apicx/checklists/[checklistId] |
| checklists | /apicx/checklists/[checklistId]/steps/[stepId]/reading |
| reports | /apicx/reports/[reportId] |
| test-records | /apicx/test-records/[testRecordId]/report/approve |
| test-records | /apicx/test-records/[testRecordId]/report |
| [versionId] | /apidocument-versions/[versionId]/assess-change |
| [documentId] | /apidocuments/[documentId]/revisions |
| [evidenceId] | /apievidence/[evidenceId]/review |
| [findingId] | /apifindings/[findingId] |
| [gateId] | /apigates/[gateId]/decisions |
| root | /apihealth |
| [jobId] | /apijobs/[jobId] |
| [...key] | /apiobjects/[...key] |
| root | /apiprofile |
| [projectId] | /apiprojects/[projectId]/activate |
| [projectId] | /apiprojects/[projectId]/alerts/[alertId] |

| Group | Endpoint |
|---|---|
| [projectId] | /apiprojects/[projectId]/alerts |
| [projectId] | /apiprojects/[projectId]/assets |
| [projectId] | /apiprojects/[projectId]/audit/verify |
| [projectId] | /apiprojects/[projectId]/claims |
| [projectId] | /apiprojects/[projectId]/compliance/checks |
| [projectId] | /apiprojects/[projectId]/compliance/precedents |
| [projectId] | /apiprojects/[projectId]/compliance/scan |
| [projectId] | /apiprojects/[projectId]/cx/checklists/[checklistId]/steps/[stepId]/vision-reading |
| [projectId] | /apiprojects/[projectId]/cx/checklists |
| [projectId] | /apiprojects/[projectId]/cx/standards |
| [projectId] | /apiprojects/[projectId]/edges |
| [projectId] | /apiprojects/[projectId]/entropy |
| [projectId] | /apiprojects/[projectId]/evidence |
| [projectId] | /apiprojects/[projectId]/export |
| [projectId] | /apiprojects/[projectId]/field-captures |
| [projectId] | /apiprojects/[projectId]/financial-model |
| [projectId] | /apiprojects/[projectId]/findings |
| [projectId] | /apiprojects/[projectId]/gates/[gateId]/readiness |
| [projectId] | /apiprojects/[projectId]/gates |
| [projectId] | /apiprojects/[projectId]/graph/nodes/[nodeId] |
| [projectId] | /apiprojects/[projectId]/graph |
| [projectId] | /apiprojects/[projectId]/knowledge/query |
| [projectId] | /apiprojects/[projectId]/knowledge/rfi-similar |
| [projectId] | /apiprojects/[projectId]/members/[memberId] |
| [projectId] | /apiprojects/[projectId]/members |
| [projectId] | /apiprojects/[projectId]/objects/[objectId]/url |
| [projectId] | /apiprojects/[projectId]/overview |
| [projectId] | /apiprojects/[projectId]/rack-models/[modelId]/equipment |
| [projectId] | /apiprojects/[projectId]/rack-models/[modelId]/export |
| [projectId] | /apiprojects/[projectId]/rack-models/[modelId]/racks |
| [projectId] | /apiprojects/[projectId]/rack-models/[modelId] |
| [projectId] | /apiprojects/[projectId]/rack-models/import |
| [projectId] | /apiprojects/[projectId]/rack-models |
| [projectId] | /apiprojects/[projectId]/readiness/recompute |
| [projectId] | /apiprojects/[projectId] |
| [projectId] | /apiprojects/[projectId]/schedule/baseline |
| [projectId] | /apiprojects/[projectId]/schedule/current |

| Group | Endpoint |
|---|---|
| [projectId] | /apiprojects/[projectId]/schedule/dependencies |
| [projectId] | /apiprojects/[projectId]/schedule/live-events |
| [projectId] | /apiprojects/[projectId]/schedule/resources |
| [projectId] | /apiprojects/[projectId]/schedule/risks |
| [projectId] | /apiprojects/[projectId]/schedule/tasks |
| [projectId] | /apiprojects/[projectId]/schedule/versions |
| [projectId] | /apiprojects/[projectId]/shipment-plans |
| [projectId] | /apiprojects/[projectId]/shipments/bulk |
| [projectId] | /apiprojects/[projectId]/shipments |
| [projectId] | /apiprojects/[projectId]/site-analysis/cooling-analysis |
| [projectId] | /apiprojects/[projectId]/site-analysis/finalize |
| [projectId] | /apiprojects/[projectId]/site-analysis/insights |
| [projectId] | /apiprojects/[projectId]/site-analysis |
| [projectId] | /apiprojects/[projectId]/sources |
| [projectId] | /apiprojects/[projectId]/systems |
| [projectId] | /apiprojects/[projectId]/technology-drafts |
| [projectId] | /apiprojects/[projectId]/turnover-packs |
| root | /apiprojects |
| [requirementId] | /apirequirements/[requirementId]/review |
| events | /apischedule/events |
| resources | /apischedule/resources/[resourceId]/review |
| risks | /apischedule/risks/[riskId]/review |
| risks | /apischedule/risks/recurring |
| tasks | /apischedule/tasks/[taskId]/review |
| versions | /apischedule/versions/[versionId]/diff |
| versions | /apischedule/versions/[versionId]/explanation |
| versions | /apischedule/versions/[versionId] |
| [shipmentId] | /apishipments/[shipmentId] |
| [packId] | /apiturnover-packs/[packId]/verify |

APPENDIX E / ENVIRONMENT AND CONFIGURATION INVENTORY

# Every example configuration key

The environment appendix is generated from the current `.env.example` file at PDF-build time. Values shown are examples only; production values must be secret-managed and individually validated.

| Key | Group | Example/default |
|---|---|---|
| DATABASE_URL | data/infrastructure | postgresql://pramana:pramana@localhost:5432/pramana |
| AUTH_MODE | identity/application | development |
| AUTH_ENCRYPTION_KEY | identity/application | replace-with-at-least-32-random-characters |
| SESSION_COOKIE_NAME | identity/application | pramana_session |
| SESSION_TTL_HOURS | identity/application | 24 |
| APP_BASE_URL | identity/application | http://localhost:3000 |
| NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY | runtime | required/optional by selected mode |
| CLERK_SECRET_KEY | identity/application | required/optional by selected mode |
| NEXT_PUBLIC_CLERK_SIGN_IN_URL | runtime | /sign-in |
| NEXT_PUBLIC_CLERK_SIGN_UP_URL | runtime | /sign-up |
| NEXT_PUBLIC_CLERK_SIGN_IN_FALLBACK_REDIRECT_URL | runtime | / |
| NEXT_PUBLIC_CLERK_SIGN_UP_FALLBACK_REDIRECT_URL | runtime | /pending-access |
| REDIS_URL | data/infrastructure | redis://localhost:6379 |
| REDIS_PREFIX | data/infrastructure | pramana |
| INFRA_ALLOW_DEGRADED | runtime | true |
| OBJECT_STORAGE_DRIVER | data/infrastructure | local |
| S3_ENDPOINT | data/infrastructure | http://localhost:9000 |
| S3_REGION | data/infrastructure | us-east-1 |
| S3_ACCESS_KEY | data/infrastructure | minioadmin |
| S3_SECRET_KEY | data/infrastructure | minioadmin |
| S3_BUCKET | data/infrastructure | pramana-local |
| MODEL_PROVIDER | AI/retrieval | ollama |
| COPILOT_MODEL_PROVIDER | runtime | ollama |
| OLLAMA_BASE_URL | AI/retrieval | http://127.0.0.1:11434 |
| OLLAMA_MODEL | AI/retrieval | gemma4:e2b |
| OLLAMA_EMBEDDING_MODEL | AI/retrieval | nomic-embed-text:latest |
| MODEL_TIMEOUT_MS | AI/retrieval | 45000 |
| MODEL_PROMPT_MAX_CHARS | AI/retrieval | 60000 |
| MODEL_OUTPUT_MAX_TOKENS | AI/retrieval | 512 |
| MODEL_CONTEXT_TOKENS | AI/retrieval | 8192 |
| GEMINI_API_KEY | AI/retrieval | required/optional by selected mode |

| Key | Group | Example/default |
|---|---|---|
| GEMINI_MODEL | AI/retrieval | gemini-2.5-flash |
| GEMINI_EMBEDDING_MODEL | AI/retrieval | text-embedding-004 |
| NIM_BASE_URL | AI/retrieval | https://integrate.api.nvidia.com/v1 |
| NIM_API_KEY | AI/retrieval | required/optional by selected mode |
| NIM_MODEL | AI/retrieval | nvidia/llama-3.3-nemotron-super-49b-v1 |
| GROQ_BASE_URL | runtime | https://api.groq.com/openai/v1 |
| GROQ_API_KEY | runtime | required/optional by selected mode |
| GROQ_MODEL | runtime | openai/gpt-oss-20b |
| CEREBRAS_BASE_URL | runtime | https://api.cerebras.ai/v1 |
| CEREBRAS_API_KEY | runtime | required/optional by selected mode |
| CEREBRAS_MODEL | runtime | gemma-4-31b |
| EMBEDDING_PROVIDER | AI/retrieval | ollama |
| PINECONE_API_KEY | runtime | required/optional by selected mode |
| PINECONE_BASE_URL | runtime | https://api.pinecone.io |
| PINECONE_API_VERSION | runtime | 2026-04 |
| PINECONE_EMBEDDING_MODEL | runtime | llama-text-embed-v2 |
| PINECONE_EMBEDDING_DIMENSIONS | runtime | 768 |
| RETRIEVAL_SERVICE_URL | AI/retrieval | http://localhost:8003 |
| KNOWLEDGE_RERANK_THRESHOLD | runtime | 0.5 |
| KNOWLEDGE_SIMILARITY_THRESHOLD | runtime | 0.5 |
| INGESTION_SERVICE_URL | domain/targets | http://localhost:8001 |
| SOLVER_SERVICE_URL | domain/targets | http://localhost:8002 |
| LOCAL_UPLOAD_DIR | data/infrastructure | ./data/uploads |
| SHIPMENT_STATUS_BUFFER_HOURS | domain/targets | 72 |
| RISK_POLL_MODE | domain/targets | synthetic |
| WEATHER_MODE | runtime | open-meteo |
| RISK_PROBABILITY_THRESHOLD | domain/targets | 0.5 |
| RISK_DELAY_HOURS_THRESHOLD | domain/targets | 8 |
| COMPLIANCE_DEVIATION_ACCURACY_TARGET | domain/targets | 0.95 |
| RISK_LEAD_TIME_TARGET_HOURS | domain/targets | 48 |
| RFI_MATCH_ACCURACY_TARGET | domain/targets | 0.85 |
| HIGH_SEVERITY_PRECISION_TARGET | domain/targets | 0.9 |
| PACK_PREP_TIME_REDUCTION_TARGET | domain/targets | 0.6 |
| SOLVER_TIMEOUT_MS | domain/targets | 90000 |
| SOLVER_MAX_ATTEMPTS | domain/targets | 3 |
| AUTH_RATE_LIMIT | identity/application | 10 |
| AUTH_RATE_LIMIT_WINDOW_SECONDS | identity/application | 60 |

| Key | Group | Example/default |
|---|---|---|
| UPLOAD_RATE_LIMIT | runtime | 20 |
| UPLOAD_RATE_LIMIT_WINDOW_SECONDS | runtime | 60 |
| SCHEDULE_RATE_LIMIT | runtime | 30 |
| SCHEDULE_RATE_LIMIT_WINDOW_SECONDS | runtime | 60 |
| EXPORT_RATE_LIMIT | runtime | 10 |
| EXPORT_RATE_LIMIT_WINDOW_SECONDS | runtime | 60 |

APPENDIX F / VERIFICATION COMMAND INVENTORY

# Every verification script currently declared

The complete test command catalog below is generated from `package.json`. It makes the release matrix auditable and prevents a single green build from being mistaken for end-to-end verification.

| Command | Runner |
|---|---|
| verify:audit | tsx scripts/verify-audit-chain.ts |
| verify:audit-concurrency | tsx scripts/verify-audit-concurrency.ts |
| verify:phase0 | tsx scripts/verify-phase0.ts |
| verify:credentials-http | tsx scripts/verify-credentials-http.ts |
| verify:schedule-http | tsx scripts/verify-schedule-http.ts |
| verify:cx-http | tsx scripts/verify-cx-http.ts |
| verify:compliance-http | tsx scripts/verify-compliance-http.ts |
| verify:risk-http | tsx scripts/verify-risk-http.ts |
| verify:poll-http | tsx scripts/verify-poll-http.ts |
| verify:risk-autopoll-http | tsx scripts/verify-risk-autopoll-http.ts |
| verify:supply-poll-http | tsx scripts/verify-supply-poll-http.ts |
| verify:weather-poll-http | tsx scripts/verify-weather-poll-http.ts |
| verify:risk-http-clients | tsx scripts/verify-risk-http-clients.ts |
| verify:knowledge-embed | tsx scripts/verify-knowledge-embed.ts |
| verify:knowledge-query-http | tsx scripts/verify-knowledge-query-http.ts |
| verify:command-links-http | tsx scripts/verify-command-links-http.ts |
| verify:rfi-similar-http | tsx scripts/verify-rfi-similar-http.ts |
| verify:evidence-turnover-http | tsx scripts/verify-evidence-turnover-http.ts |
| verify:graph-expansion-http | tsx scripts/verify-graph-expansion-http.ts |
| verify:live-events-http | tsx scripts/verify-live-events-http.ts |
| verify:turnover-cx-http | tsx scripts/verify-turnover-cx-http.ts |
| verify:gate-context-http | tsx scripts/verify-gate-context-http.ts |
| verify:change-impact-http | tsx scripts/verify-change-impact-http.ts |
| verify:hardening-http | tsx scripts/verify-hardening-http.ts |
| verify:polling-http | tsx scripts/verify-polling-http.ts |
| verify:knowledge-http | tsx scripts/verify-knowledge-http.ts |
| verify:model-provider | tsx scripts/verify-model-provider.ts |
| verify:rack-model | tsx scripts/verify-rack-model.ts |
| verify:provider-safety | tsx scripts/verify-provider-safety.ts |
| verify:ollama | tsx scripts/verify-ollama-provider.ts |
| verify:deep-links | tsx scripts/verify-deep-links.ts |
| verify:data-integrity | tsx scripts/verify-data-integrity.ts |

| Command | Runner |
|---|---|
| verify:live-providers | tsx scripts/verify-live-providers-fail-closed.ts |
| verify:retrieval-service | tsx scripts/verify-retrieval-service.ts |
| verify:knowledge-rerank | tsx scripts/verify-knowledge-rerank-http.ts |
| verify:knowledge-synthesis | tsx scripts/verify-knowledge-synthesis-http.ts |
| verify:compliance-scan-http | tsx scripts/verify-compliance-scan-http.ts |
| verify:compliance-llm-http | tsx scripts/verify-compliance-llm-http.ts |
| verify:risk-mitigations-http | tsx scripts/verify-risk-mitigations-http.ts |
| verify:ingestion-formats-http | tsx scripts/verify-ingestion-formats-http.ts |
| verify:compliance-modality-http | tsx scripts/verify-compliance-modality-http.ts |
| verify:turnover-provenance-http | tsx scripts/verify-turnover-provenance-http.ts |
| verify:config-targets | tsx scripts/verify-config-targets.ts |
| verify:solver-resilience | tsx scripts/verify-solver-resilience.ts |
| verify:rate-limit-coverage | tsx scripts/verify-rate-limit-coverage.ts |
| verify:compliance-finding-fields | tsx scripts/verify-compliance-finding-fields.ts |
| verify:compliance-golden | tsx scripts/verify-compliance-golden.ts |
| verify:compliance-authority | tsx scripts/verify-compliance-authority.ts |
| verify:overdue-findings | tsx scripts/verify-overdue-findings.ts |
| verify:turnover-schedule-provenance | tsx scripts/verify-turnover-schedule-provenance.ts |
| verify:evidence-entropy | tsx scripts/verify-evidence-entropy.ts |
| verify:teachback-http | tsx scripts/verify-teachback-http.ts |
| verify:rfi-resolution | tsx scripts/verify-rfi-resolution.ts |
| verify:knowledge-filters | tsx scripts/verify-knowledge-filters.ts |
| verify:copilot | tsx scripts/verify-copilot-http.ts |
| verify:all | tsx scripts/verify-all.ts |

APPENDIX G / FULL TOP-20 DEFECT AND RESOLUTION LEDGER

# Complete merged-branch remediation record

| # | Defect | Resolution |
|---|---|---|
| 01 | Mock output could look like a real LLM. | Explicit real provider configuration, Ollama default, production mock rejection and provenance. |
| 02 | AI paths bypassed provider architecture. | Shared schema-validated provider boundary across generation, vision, compliance, risk and knowledge. |
| 03 | Generation and embeddings could select contradictory providers. | Independent explicit provider configuration plus embedding-model tagging and mismatch exclusion. |
| 04 | Long model work could freeze the frontend. | Prompt/context/token limits, server deadline, client abort, progress and bounded retry. |
| 05 | Deep links opened a generic page. | Query-ID consumption/validation and focused destination record state. |
| 06 | Duplicate React keys could omit or duplicate cards. | Authoritative-id dedupe and composite stable UI keys. |
| 07 | Duplicate systems/gates/edges/chunks/alerts were possible. | Business-key constraints, seed reconciliation and integrity verifier. |
| 08 | Two branch migration sequences conflicted. | Final schema reconciliation, ordered migration history and replay validation. |
| 09 | Processed PDFs could be absent from RAG. | Extraction-to-indexing contract and historical semantic-chunk backfill. |
| 10 | RAG could attribute another standard to named document. | Document selection, exact title routing and SQL filtering before ranking. |
| 11 | LLM planner could narrow authority scope incorrectly. | Only explicit filters/deterministic title resolution may narrow metadata. |
| 12 | Fallback text could masquerade as grounded synthesis. | Structured cited synthesis or explicit no-results; unsupported claims dropped. |
| 13 | Cx could mix cross-scope gate/system/asset. | Project/system relationship validation and 422 rejection. |
| 14 | Raw hashes/JSON/placeholder characters leaked into UI. | Presentation helpers, labels, bounded precision and contextual provenance links. |
| 15 | Long operations had inconsistent feedback. | Per-operation busy state, status messages and disabled conflicting controls. |
| 16 | UI navigation was crowded and weakly structured. | Task-oriented groups, responsive navigation and contextual actions. |
| 17 | Shipment map could lack route/position context. | Saved-coordinate routes, markers, fit bounds and explicit no-coordinate state. |
| 18 | Land-routing fallback could mislead. | HTTPS/deadline routing, mode-aware provenance and safe fallback labels. |
| 19 | Synthetic signals could look operational. | Live/synthetic/unavailable provenance and fail-closed client behaviour. |
| 20 | No trustworthy release gate existed. | Full verification matrix, audit/dependency checks and explicit production prerequisites. |

## Source of truth

The complete narrative, original branch attribution and verification references remain in README.md under 'Top 20 defects found across Refinement and Updated-Refinement'. This appendix preserves every remedial category in the printable architecture reference.

APPENDIX H / FEATURE AND USER-STORY COVERAGE

# Full planned-to-implemented product map

| Stories | Capability group | Controlled outcome |
|---|---|---|
| US-01 to US-08 | Project access, source ingestion, requirement review, readiness, actions, change impact, decisions and turnover | A project-scoped evidence trail from controlled source to approved, verifiable pack. |
| US-09 to US-12 | Baseline schedule, event-triggered rescheduling, explanation and gate schedule context | Accepted task/resource inputs lead to immutable CP-SAT versions and cross-linked context, not automatic readiness changes. |
| US-13 to US-18 | Standards checklist, IST execution, deterministic checks, failures, reports and evidence linkage | Cited Cx workflow keeps numerical/boolean verdicts deterministic and narrative judgement human-routed. |
| US-19 to US-23 | Shipment tracking, weather ETA, risk status, map navigator and schedule events | Asset-linked visibility with explicit provenance and delta-gated schedule influence. |
| US-24 to US-25 | Specification compliance and approved-equal grounding | Proposed deviations are exact-citation grounded and require human disposition. |
| US-26 to US-27 | Predictive risk and live-event surfaces | Advisory material risks cross-link critical path but never autonomously alter a plan. |
| US-28 to US-31 | Knowledge query, similar RFI, project graph/timeline and unified Command Center | Project-scoped cited intelligence, typed graph navigation and deduplicated operational alerts. |

## Explicit product boundaries

- No general cross-project chatbot, autonomous commissioning authority, independent certification, auto-closure of NCRs/waivers/tests or AI-issued gate state.
- No hidden substitution of synthetic AIS/weather/model output for verified live operations.
- No unlicensed standards/client/vendor content in reusable demos, prompts, embeddings or training material.
- No second authoritative graph/vector store: Postgres/pgvector/edges remain authority; service-local working state cannot override it.